PRISMSPACE - INTELLIGENT MULTI-AGENT PLATFORM



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Project Group 01

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01. Introduction: PrismSpace Web Platform

PrismSpace is an interactive, **web-based multi-agent intelligence platform** and workspace website that allows users to orchestrate autonomous AI swarms directly from modern web browsers. It converts natural-language user goals into dependency-aware **Directed Acyclic Graphs (DAGs)** for parallel sub-task execution. Built as a responsive web application (Next.js 15, React 19) accessible from any browser with zero local setup, it features an interactive visual DAG swarm visualizer with real-time token and step streaming. The platform pairs Groq, OpenAI, and NVIDIA NIM with standardized tool connectivity via Model Context Protocol (MCP), while storing all session state locally in client-side IndexedDB (`PrismDB`) with human-in-the-loop approval gates for privacy and safety.

Problem Statement

Challenges in current AI-agent and workflow systems

02. Problem Statement: Core Challenges in AI Agent Systems

Current AI-agent systems suffer from monolithic model bottlenecks, rigid sequential execution, fragmented tool integrations, limited observability, hallucinated tool calls, and unsafe or privacy-sensitive actions. **PrismSpace** takes developer task prompts, workspace and session state, uploaded files, and MCP/HITL configurations as **inputs**; applies intent classification, supervised model/provider routing, DAG decomposition with topological scheduling, MCP-enabled ReAct execution, ORPO preference alignment, and human approval as its **method**; and produces **outputs** including verified artifacts, tool results, execution traces, quality and cost telemetry, and persistent session records.

03. Objectives

1. Decompose complex user goals into dependency-aware **DAGs** for structured and parallel task execution.
2. Orchestrate specialist agents and approved tools through **MCP** and bounded **ReAct** loops.
3. Align generated responses using **ORPO** preference optimization for quality, faithfulness, and safety.
4. Route tasks to suitable models and providers using supervised capability, cost, and latency signals.
5. Validate risky actions with **HITL** approval and preserve verified outputs with execution telemetry.

04. Literature Review

Research background, key findings, and identified gap

Multi-Objective Reward Generalization in Deep Reinforcement Learning

Citation	Summary	Advantages & Disadvantages
F. Cornalba, C. Disselkamp, D. Scassola, and C. Helf *Neural Computing and Applications* **Year:** 2024 **Publisher:** Springer Open Access	Investigates **Multi-Objective Deep Reinforcement Learning (MODRL)** for optimizing competing operational objectives (e.g., return vs. risk). Introduces reward generalization techniques that train pareto-optimal policies capable of dynamically adapting to shifting stakeholder preferences without retraining from scratch.	**Advantages:** ▶ Provides structured Pareto-optimal policy exploration across conflicting objectives. ▶ Generalizes across parameterized reward weights without full policy retraining. ▶ Directly informs multi-objective routing (latency vs. cost vs. quality) in PrismSpace. **Disadvantages:** ▶ High initial training sample complexity across multi-dimensional reward spaces. ▶ Evaluated primarily on continuous financial time-series data. ▶ Sensitive to hyperparameter tuning in policy entropy regularization.

Average Reward Adjusted Discounted Reinforcement Learning

Citation	Summary	Advantages & Disadvantages
M. Schneckenreither and G. Moser *Neural Computing and Applications* **Year:** 2025 **Publisher:** Springer Open Access	Proposes **Average-Reward Adjusted Discounted RL (Reinforcement Learning) (ARAD-RL)**, bridging the gap between discounted and average reward criteria in infinite-horizon **MDPs** (Markov Decision Processes). Addresses systematic value estimation bias in standard discounted Q-learning by dynamically subtracting the estimated gain rate during Bellman updates.	**Advantages:** ▶ Eliminates infinite-horizon return explosion while preserving policy convergence. ▶ Improves long-term steady-state throughput in continuous scheduling tasks. ▶ Theoretically sound formulation for continuous agent lifecycle optimization. **Disadvantages:** ▶ Requires accurate empirical estimation of average reward baseline. ▶ Slow convergence in highly non-stationary transition environments. ▶ Adds variance to early-stage **TD** (Temporal Difference) error gradient estimates.

Deep Reinforcement Learning for Multi-Agent Coordination

Citation	Summary	Advantages & Disadvantages
K. Aina and S. Ha *Artificial Life and Robotics* **Year:** 2026 **Publisher:** Springer Open Access	Develops decentralized **MARL** (Multi-Agent Reinforcement Learning) algorithms for multi-agent coordination in constrained environments. Employs **spatial-temporal observation encoding** and localized interaction graphs to coordinate joint movements and resolve deadlocks without requiring centralized global state knowledge.	**Advantages:** ▶ Resolves shared-resource contention and deadlocks via decentralized policies. ▶ Spatial-temporal graph attention scales linearly with agent density. ▶ Applicable to concurrent tool-use contention across worker agents. **Disadvantages:** ▶ Tested in physical robotics grids rather than software **API** (Application Programming Interface) environments. ▶ Vulnerable to communication packet dropouts and sensor latency. ▶ Does not account for semantic language-based coordination.

Open-Ended Coordination Using Modular Open Policies

Citation	Summary	Advantages & Disadvantages
D. Rother, J. Pajarinen, J. Peters, and T. H. Weisswange *Autonomous Agents and Multi-Agent Systems* **Year:** 2025 **Publisher:** Springer Open Access	Presents a framework for **open-ended ad hoc multi-agent coordination**. Utilizes **Modular Open Policies (MOP)** where dynamic agent teams dynamically infer unseen teammate skill representations and synthesize blended coordination strategies without requiring pre-coordinated joint training.	**Advantages:** ▶ Enables zero-shot ad hoc coordination with unfamiliar teammate agent policies. ▶ Modular policy decomposition allows plug-and-play agent role assignment. ▶ Directly adopted in PrismSpace for dynamic worker swarm scaling. **Disadvantages:** ▶ Requires comprehensive pre-trained skill policy libraries for base roles. ▶ Evaluated primarily on benchmark games (Overcooked, **LBF** (Level-Based Foraging)). ▶ Action blending may introduce transient sub-optimal execution steps.

Improving Scalability of **MADRL** (Multi-Agent Deep Reinforcement Learning) with Suboptimal Human Knowledge

Citation	Summary	Advantages & Disadvantages
D. Liu, F. Ren, J. Yan, G. Su, W. Gu, and S. Kato *Autonomous Agents and Multi-Agent Systems* **Year:** 2026 **Publisher:** Springer Open Access	Addresses sample inefficiency in large-scale **Multi-Agent DRL** (Deep Reinforcement Learning) by integrating heuristic human guidance into exploration. Proposes an adaptive filtering mechanism that extracts actionable macro-actions from noisy or sub-optimal human demonstration traces, accelerating multi-agent policy learning.	**Advantages:** ▶ Accelerates policy exploration by up to 4x using noisy demonstration priors. ▶ Robust to suboptimal or conflicting human heuristic rules. ▶ Informs PrismSpace's bootstrap initialization from logged developer sessions. **Disadvantages:** ▶ Quality of learned policies depends on initial heuristic coverage. ▶ Requires complex threshold tuning to transition away from demonstrations. ▶ High memory footprint for maintaining dual replay buffers.

ε-Retraining Reinforcement Learning Algorithms

Citation	Summary	Advantages & Disadvantages
L. Marzari, C. Liu, P. L. Donti, and E. Marchesini *Autonomous Agents and Multi-Agent Systems* **Year:** 2026 **Publisher:** Springer Open Access	Investigates bounded policy fine-tuning through ε-**retraining**, preventing catastrophic forgetting when deep **RL** agents adapt to non-stationary environments. Establishes formal Lipschitz bounds on policy drift, ensuring safe continual learning in deployed agent pipelines.	**Advantages:** ▶ Provides mathematical guarantees against catastrophic policy divergence. ▶ Enables continuous telemetry-driven model fine-tuning in production. ▶ Crucial for stability in PrismSpace's ongoing user interaction learning. **Disadvantages:** ▶ Constrained optimization increases computation per gradient step. ▶ Conservative update bounds may slow adaptation to drastic distribution shifts. ▶ Theoretical guarantees rely on local smoothness assumptions.

Deep Multiagent Reinforcement Learning: Challenges and Directions

Citation	Summary	Advantages & Disadvantages
A. Wong, T. Bäck, A. V. Kononova, and A. Plaat *Artificial Intelligence Review* **Year:** 2023 **Publisher:** Springer Open Access	Surveys **Deep Multiagent Reinforcement Learning (MARL)**, formalizing foundational challenges including non-stationarity, multi-agent credit assignment, and combinatorial action explosion. Evaluates **Centralized Training with Decentralized Execution (CTDE)** and communication architectures for modular orchestration.	**Advantages:** ▶ Comprehensive taxonomy of credit assignment and **CTDE** paradigms. ▶ Establishes formal theoretical criteria for communication protocols. ▶ Provides the theoretical foundation for PrismSpace's modular agent architecture. **Disadvantages:** ▶ Survey paper without empirical experimental evaluation. ▶ Focuses on discrete game simulations rather than **API**-driven software tasks. ▶ Does not cover **LLM** reasoning agents or human-in-the-loop approval gates.

HIAT (Human-in-the-Loop with Auxiliary Tasks): Human-in-the-Loop **RL** with Auxiliary Tasks

Citation	Summary	Advantages & Disadvantages
B. Niu, B. Luo, Y. Cui, X. Xu, Y. Zhao, and Y. Feng *Artificial Intelligence Review* **Year:** 2025 **Publisher:** Springer Open Access	Introduces **HIAT (Human-in-the-Loop with Auxiliary Tasks)**, a framework leveraging human intervention signals as auxiliary self-supervised learning objectives. Models human corrections as feedback gradients that shape internal feature representations, minimizing human intervention frequency over time.	**Advantages:** ▶ Maximizes learning efficiency from scarce human intervention signals. ▶ Significantly reduces human cognitive load through auxiliary reward shaping. ▶ Informs PrismSpace's approval gate telemetry and learning from user edits. **Disadvantages:** ▶ Requires consistent human feedback to avoid biased auxiliary representations. ▶ Intervention timing latency can cause temporal mis-alignment in **RL** updates. ▶ Higher computational overhead during auxiliary task representation training.

ORPO (Odds Ratio Preference Optimization): Monolithic Preference Optimization without Reference Model

Citation	Summary	Advantages & Disadvantages
J. Hong, N. Lee, and J. Thorne *EMNLP* / *arXiv:2403.07691* **Year:** 2024 **Area: LLM** Preference Alignment	**Core Idea (Single-Phase Alignment):** Instead of multi-step **RLHF**, **ORPO** teaches desired answers while penalizing bad answers in one unified equation: $$\mathcal{L}_{\text{ORPO}} = \mathcal{L}_{\text{SFT}} + \lambda \cdot \mathcal{L}_{\text{OR}}$$ $$\mathcal{L}_{\text{OR}} = -\log \sigma \left(\log \frac{\text{odds}(y_w \mid x)}{\text{odds}(y_l \mid x)} \right)$$ **Key Terms:** ▶ $\mathcal{L}_{\text{SFT}}$: Base supervised fine-tuning loss (correct code/syntax). ▶ $\mathcal{L}_{\text{OR}}, \lambda$: Odds ratio penalty term and penalty weight. ▶ x, y_w, y_l: Prompt, winning action path, and losing action path.	**Advantages:** ▶ Reference-model-free: eliminates maintaining frozen π_{ref} in memory. ▶ Monolithic single-phase training avoids unstable **PPO** policy optimization. ▶ Directly aligns PrismSpace agent trajectories on paired feedback (*chosen* vs. *rejected*). **Disadvantages:** ▶ Requires high-quality pairwise preference datasets (y_w, y_l). ▶ Sensitive to λ hyperparameter balance.

Reinforcement Online Learning to Rank with Unbiased Reward Shaping

Citation	Summary	Advantages & Disadvantages
S. Zhuang, Z. Qiao, and G. Zuccon *Information Retrieval Journal* **Year:** 2023 **Publisher:** Springer Open Access	Develops an **Online Learning to Rank (OLTR)** framework using counterfactual reinforcement learning. Employs **inverse propensity weighting (IPW)** and unbiased reward shaping to correct for position, presentation, and click bias in user interaction logs during continuous ranker optimization.	**Advantages:** ▶ Guarantees unbiased gradient estimation from observational implicit feedback. ▶ Operates entirely online without requiring manual document relevance labels. ▶ Directly informs PrismSpace's context retrieval ranking feedback loop. **Disadvantages:** ▶ High propensity score variance can destabilize gradient updates. ▶ Requires large volumes of query-click logs to converge. ▶ Assumes position-based user click models that may not hold in code search.

Knowledge Learning for Securing Workflow Scheduling in **MEC** (Mobile Edge Computing)

Citation	Summary	Advantages & Disadvantages
T. M. Ghazal, A. E. Awouda, M. K. Hasan, A. H. A. Rahman, S. Islam, R. A. Saeed, H. Elshafie, A. Iqbal, and A. H. Hussein *Mobile Networks and Applications* **Year:** 2025 **Publisher:** Springer Open Access	Proposes a **Knowledge-Guided Reinforcement Learning (KG-RL)** scheduler for complex workflow **DAGs** in distributed edge-cloud environments. Integrates security constraint validation directly into the state transition model, preventing insecure task placement while optimizing makespan.	**Advantages:** ▶ Enforces strict security boundary constraints during automated task scheduling. ▶ Effectively handles multi-dependency Directed Acyclic Graph (**DAG**) workflows. ▶ Informs PrismSpace's sandboxed tool execution and dependency resolution. **Disadvantages:** ▶ Constraint checking introduces scheduling latency for massive **DAGs**. ▶ Assumes static security classification matrices across network nodes. ▶ Limited evaluation on non-deterministic tool failure scenarios.

Energy-Aware Cooperative Multi-Fitness Evolutionary Workflow Scheduling

Citation	Summary	Advantages & Disadvantages
P. Barredo and J. Puente *Natural Computing* **Year:** 2025 **Publisher:** Springer Open Access	Presents a **Cooperative Multi-Fitness Evolutionary Algorithm (CMFEA)** for multi-objective workflow scheduling. Optimizes trade-offs between execution time, monetary compute cost, and energy consumption using island-model cooperative coevolution across heterogeneous worker pools.	**Advantages:** ▶ Pareto-optimal trade-offs across makespan, financial cost, and energy. ▶ Cooperative sub-population decomposition prevents premature local convergence. ▶ Provides heuristics for PrismSpace's multi-agent model provider selection. **Disadvantages:** ▶ Evolutionary search has higher runtime latency than single-pass ML heuristics. ▶ Cannot adapt instantaneously to real-time **API** rate limit drops. ▶ Requires parameter tuning across mutation, crossover, and migration rates.

Optimized Task Scheduling Using Hybrid **DL** (Deep Learning) and Metaheuristics

Citation	Summary	Advantages & Disadvantages
A. Heirati, M. Fartash, and M. Khalily-Dermany *Neural Processing Letters* **Year:** 2026 **Publisher:** Springer Open Access	Combines **Deep Learning (LSTM (Long Short-Term Memory))** workload prediction with metaheuristic optimization (Harris Hawks Optimization) for proactive workflow dispatching. Predicts task execution durations and memory bottlenecks in advance to prevent node overloading and minimize queue waiting times.	**Advantages:** ▶ Proactive workload forecasting prevents sudden compute node thrashing. ▶ Hybrid pipeline balances search exploration with predictive dispatching. ▶ Relevant for predicting token latency and cost before agent invocation. **Disadvantages:** ▶ Two-stage pipeline introduces inference overhead before scheduling decisions. ▶ **LSTM** prediction accuracy degrades under sudden irregular workload spikes. ▶ Complex offline model retraining requirements.

Cost-Aware Workflow Offloading in Edge-Cloud Computing via **GA** (Genetic Algorithm)

Citation	Summary	Advantages & Disadvantages
S. Abdi, M. Ashjaei, and S. Mubeen *The Journal of Supercomputing* **Year:** 2024 **Publisher:** Springer Open Access	Proposes a **Genetic Algorithm-based workflow offloading** framework considering heterogeneous network bandwidth, deadline constraints, and monetary costs. Implements chromosome repair operators to preserve **DAG** precedence validity throughout mutation and crossover.	**Advantages:** ▶ Guarantees valid **DAG** dependency structures via specialized repair operators. ▶ Explicitly accounts for network transfer latency between pipeline stages. ▶ Informs subtask payload transfer between worker agents in PrismSpace. **Disadvantages:** ▶ Computationally heavy for interactive real-time user query response. ▶ Assumes deterministic network transfer bandwidth in cloud environments. ▶ Does not model dynamic agent failures or retries.

Dynamic Role Discovery and Assignment in Multi-Agent Task Decomposition

Citation	Summary	Advantages & Disadvantages
Y. Xia, J. Zhu, and L. Zhu *Complex & Intelligent Systems* **Year:** 2023 **Publisher:** Springer Open Access	Proposes **dynamic role discovery and assignment** that decomposes complex high-level objectives into lower-level subtasks. Uses learned persona embeddings and mutual information to adaptively allocate subtasks according to situational agent capabilities, eliminating rigid static assignments.	**Advantages:** ▶ Enables flexible subtask allocation without hardcoded role hierarchies. ▶ Hierarchical decomposition prevents single-agent cognitive context overload. ▶ Serves as core theoretical basis for PrismSpace's dynamic agent routing. **Disadvantages:** ▶ Role discovery clustering introduces upfront computational overhead. ▶ Performance depends on representation vector accuracy for subtasks. ▶ Evaluated primarily on synthetic collaborative coordination benchmarks.

Decision Variable Contribution Based Adaptive Multi-Objective Scheduling

Citation	Summary	Advantages & Disadvantages
J. Li, L. Xing, W. Zhong, Z. Cai, and F. Hou *Complex & Intelligent Systems* **Year:** 2023 **Publisher:** Springer Open Access	Introduces a **Decision Variable Contribution (DVC)** adaptive mechanism for evolutionary multi-objective workflow scheduling. Evaluates the differential sensitivity of individual decision variables to focus evolutionary search operators on high-impact scheduling bottlenecks.	**Advantages:** ▶ Accelerates Pareto-front convergence by prioritizing high-variance variables. ▶ Dynamically adapts search intensity across workflow bottleneck stages. ▶ Useful for pinpointing high-latency critical path tools in agent workflows. **Disadvantages:** ▶ Sensitivity calculation adds per-generation matrix computation cost. ▶ Can over-concentrate on dominant objectives at the expense of diversity. ▶ Requires calibrated normalization across widely different metric units.

GHQ: Grouped Hybrid Q-Learning for Heterogeneous Multi-Agent RL

Citation	Summary	Advantages & Disadvantages
X. Yu, Y. Lin, X. Wang, S. Han, and K. Lv *Complex & Intelligent Systems* **Year:** 2024 **Publisher:** Springer Open Access	Develops **Grouped Hybrid Q-learning (GHQ)** for cooperative heterogeneous multi-agent systems. Clusters agents with similar capabilities into functional cohorts that share value approximations while maintaining decoupled action execution heads for role specialization.	**Advantages:** ▶ Scales value estimation across large heterogeneous swarms via cohort grouping. ▶ Reduces joint action space dimensionality from exponential to block-linear. ▶ Directly applicable to categorizing **LLM** worker agent skill specializations. **Disadvantages:** ▶ Cohort clustering boundaries can be rigid if agent capabilities evolve. ▶ Intra-group credit assignment remains non-trivial in dense interactions. ▶ Requires domain knowledge to establish initial grouping heuristics.

Decentralized Multi-Agent Cooperation via Adaptive Partner Modeling

Citation	Summary	Advantages & Disadvantages
C. Xu, J. Wang, X. Zhu, Y. Yue, W. Zhou, Z. Liang, and D. Wojtczak *Complex & Intelligent Systems* **Year:** 2024 **Publisher:** Springer Open Access	Tackles environmental non-stationarity in decentralized multi-agent cooperation using **Adaptive Partner Modeling (APM)**. Instead of relying on expensive continuous inter-agent broadcast communication, agents infer partner policies and intended actions from local observation histories.	**Advantages:** ▶ Minimizes inter-agent communication bandwidth to near zero during execution. ▶ Stabilizes decentralized joint execution despite teammate policy shifts. ▶ Directly adopted in PrismSpace for zero-communication agent coordination. **Disadvantages:** ▶ Partner inference error propagates if historical observations are noisy. ▶ Inference networks introduce local computational and memory overhead. ▶ Assumes partners follow relatively stationary parameterized policy spaces.

A Responsible AI Framework: Pipeline Contextualisation

Citation	Summary	Advantages & Disadvantages
E. Vyhmeister, G. G. Castañé, P.-O. Östberg, and S. Thevenin *AI and Ethics* **Year:** 2023 **Publisher:** Springer Open Access	Formulates an operational **Responsible AI Framework** providing pipeline contextualisation across the full model lifecycle. Specifies auditable checkpoints, data lineage tracking, and bias mitigation protocols for autonomous AI decision-making systems.	**Advantages:** ▶ Holistic governance framework linking technical architecture to ethical safety. ▶ Defines concrete telemetry points for data provenance and action auditing. ▶ Guides PrismSpace's execution audit logging and human inspection UI. **Disadvantages:** ▶ Qualitative framework that requires custom translation into concrete code gates. ▶ Audit logging introduces storage overhead and slight execution latency. ▶ Does not provide automated formal verification of **LLM** reasoning traces.

Meaningful Human Control: Actionable Properties for AI Development

Citation	Summary	Advantages & Disadvantages
L. Cavalcante Siebert, M. L. Lupetti, E. Aizenberg, N. Beckers, A. Zgonnikov, H. Veluwenkamp, D. Abbink, E. Giaccardi, G.-J. Houben, C. M. Jonker, J. van den Hoven, D. Forster, and R. L. Lagendijk *AI and Ethics* **Year:** 2023 **Publisher:** Springer Open Access	Formalizes the philosophical concept of **Meaningful Human Control (MHC)** into actionable software design properties: **Tracking** (system responsive to human moral reasons) and **Tracing** (system actions causally attributable to accountable human supervisors).	**Advantages:** ▶ Translates abstract governance principles into rigorous software specifications. ▶ Establishes tracking and tracing requirements for autonomous tool execution. ▶ Directly guides the design of PrismSpace's Human-in-the-Loop (**HITL**) approval gates. **Disadvantages:** ▶ High human intervention frequency can degrade automated agent throughput. ▶ Defining actionable reason boundaries requires careful domain calibration. ▶ Assumes human supervisor maintains full comprehension of complex agent plans.

CPN-HRL (Cloud-Private-Network Hierarchical Reinforcement Learning): Hierarchical **DRL** for Priority-Aware Task Scheduling

Citation	Summary	Advantages & Disadvantages
Q. Yue, L. Tian, X. Feng, Z. Yuan, S. Wei, J. Xia, B. Chen, X. Song, and Y. Yao *Journal of King Saud University CIS* **Year:** 2026 **Publisher:** Springer Open Access	Presents **CPN-HRL**, a Hierarchical Reinforcement Learning framework for priority-aware task scheduling in Cloud-Edge-End networks. Uses a macro-policy to partition task **DAGs** by priority and latency budgets, while micro-policies schedule subtasks onto optimal compute nodes.	**Advantages:** ▶ Hierarchical policy decomposition effectively handles complex task **DAGs**. ▶ Guarantees priority-aware quality of service under tight latency deadlines. ▶ Serves as primary reference for PrismSpace's **DAG** workflow scheduling engine. **Disadvantages:** ▶ Multi-tier policy architecture requires synchronous training coordination. ▶ Edge node failure requires immediate micro-policy replanning. ▶ Evaluated in simulated telecom cloud-edge topologies.

Automatic Discovery and Description of Human Planning Strategies

Citation	Summary	Advantages & Disadvantages
J. Skirzyński, Y. Jain, and F. Lieder *Behavior Research Methods* **Year:** 2023 **Publisher:** Springer Open Access	Develops an algorithmic framework for **automatically discovering and formalizing planning strategies** from human behavior traces. Models task planning as resource-rational search over Directed Graphical Models, producing interpretable decision trees that characterize human heuristic planning.	**Advantages:** ▶ Extracts interpretable, human-like heuristic planning strategies automatically. ▶ Quantifies trade-offs between cognitive search effort and decision quality. ▶ Informs PrismSpace's hierarchical prompt decomposition and agent planning. **Disadvantages:** ▶ Search space for strategy discovery grows exponentially with problem depth. ▶ Human behavioral experiments conducted in stylized cognitive testbeds. ▶ Static decision trees struggle to capture fluid **LLM** reasoning trajectories.

Data-Intensive Workflow Scheduling Based on **DRL** in Multi-Clouds

Citation	Summary	Advantages & Disadvantages
S. Zhang, Z. Zhao, C. Liu, and S. Qin *Journal of Cloud Computing* **Year:** 2023 **Publisher:** Springer Open Access	Proposes a **Deep Q-Network (DQN)** scheduler optimized for data-intensive workflows across multi-cloud environments. Explicitly models dataset placement and cross-datacenter transfer latency to minimize both workflow completion time and data egress billing.	**Advantages:** ▶ Optimizes data locality to prevent massive cross-cloud egress overhead. ▶ Adaptive reward function balances monetary cost and makespan constraints. ▶ Directly relevant to caching and embedding retrieval across distributed agents. **Disadvantages:** ▶ Assumes deterministic file sizes and network bandwidth availability. ▶ Q-network value overestimation requires double-**DQN** correction. ▶ State representation size scales with the total number of cloud datacenters.

RL-Based Task Scheduling for Sustainable Federated Cloud Computing

Citation	Summary	Advantages & Disadvantages
Z. Wang, S. Chen, L. Bai, J. Gao, J. Tao, R. R. Bond, and M. D. Mulvenna *Journal of Cloud Computing* **Year:** 2023 **Publisher:** Springer Open Access	Develops an **Actor-Critic RL scheduling framework** for federated cloud datacenters that incorporates renewable energy availability into scheduling policies. Dynamically routes compute tasks to datacenters with green energy surpluses without violating user SLA deadlines.	**Advantages:** ▶ Incorporates environmental sustainability and carbon metrics into scheduling. ▶ Actor-Critic policy gradient handles continuous action spaces effectively. ▶ Provides conceptual foundation for multi-attribute provider routing. **Disadvantages:** ▶ Requires real-time telemetry from green energy generation sources. ▶ Trade-off between carbon footprint and network transmission latency. ▶ Sensitive to reward scaling factors between SLA penalties and green rewards.

Hybrid Dolphin Swarm Sparrow Search Multi-Objective Workflow Scheduling

Citation	Summary	Advantages & Disadvantages
S. Subashree, M. Rajakumaran, G. Pushpa, and S. Senthilkumar *Journal of Cloud Computing* **Year:** 2025 **Publisher:** Springer Open Access	Proposes a hybrid bio-inspired metaheuristic combining **Dolphin Swarm Optimization (DSA)** and **Sparrow Search Algorithm (SSA)** for multi-objective cloud workflow scheduling. Optimizes execution makespan, virtual machine utilization, and monetary execution expenditure.	**Advantages:** ▶ Combines global exploration of DSA with local exploitation of SSA. ▶ Demonstrates superior Pareto hypervolume compared to standard **GA** and **PSO** (Particle Swarm Optimization). ▶ Provides strong comparative baselines for workflow optimization algorithms. **Disadvantages:** ▶ Metaheuristic iteration runtime is too slow for sub-second reactive routing. ▶ Multiple tuning parameters (predation rate, vocalization index) required. ▶ Lacks theoretical convergence rate bounds for stochastic optimization.

Research Gap

What the surveyed methods already do

MARL works address **CTDE**, role grouping, and ad-hoc coordination. **ORPO** aligns **LLM** preferences without reference models, while **HIAT** leverages human interventions as auxiliary **RL** signals. **CPN-HRL** and **DAG** schedulers optimize cloud task allocation, and **MHC** formalizes tracking and tracing properties.

Limitations observed in those papers

Coordination benchmarks focus on gaming (**SMAC**, Overcooked) rather than filesystem/**API** tooling. **ORPO** lacks grounding in **MCP ReAct** tool traces, while **HIAT** targets continuous robotic control instead of code execution. Cloud schedulers assume static **DAGs** or slow metaheuristics ill-suited for real-time **LLM** routing.

Why PrismSpace is required

No reviewed system jointly grounds tool observations, aligns policies on agent trajectories, dynamically routes heterogeneous **LLMs**, and enforces human-in-the-loop safety gates within a unified developer workspace.

03. Comparative Analysis: Existing Platforms vs. PrismSpace

Simple comparison: Comparing PrismSpace with conversational AI, autonomous-agent, and multi-agent platforms.

Capability Dimension	ChatGPT Web (OpenAI)	AutoGPT (Autonomous Loop)	CrewAI (Multi-Agent Swarm)	PrismSpace (Ours) Local Multi-Agent Platform
Execution Topology	One conversation at a time	Repeats steps automatically; may not stop	Agents follow a fixed sequence	**DAG (Directed Acyclic Graph) Scheduling:** Runs dependent tasks in parallel, with up to three workers
Model Strategy & Routing	Mostly one provider and model	Usually one model with custom scripts	Model selected for each agent role	**Multi-Model Router:** Selects a suitable provider using task needs, speed, and cost
Tool Connectivity Protocol	Built-in plugins and function calls	Custom scripts and shell commands	Tool wrappers with fixed formats	**MCP (Model Context Protocol):** A common interface for tools such as Gmail, Drive, files, and web services
Observability & Streaming	Shows answer text; internal steps are hidden	Terminal logs only	Text traces from agent execution	**Live Run Status + Workflow Graph:** Shows task progress and each node's state
Safety & Quality Control	Provider filters; no user approval step	Fixed limits; limited output checking	Agent reviews its own output	**HITL (Human-in-the-Loop) + ORPO (Odds Ratio Preference Optimization):** Pauses risky tasks and ranks preferred outputs
Persistence & Privacy	Data stored in the provider's cloud	Local files without a structured database	Cloud or temporary in-memory state	**Local-First PrismDB:** Stores sessions and files in the browser's IndexedDB database

PrismSpace summary: It combines dependency-aware task planning, shared tool connections, provider selection, visible progress, and local browser storage in one workspace.

Algorithm

Step-by-step execution of the PrismSpace multi-agent workflow

Algorithm: Step 1 – Understand the User Prompt

Step 1: Prompt Parsing and Structured Representation

1.1 Read the user's natural-language prompt and identify the main goal.

1.2 Retrieve persistent session context and workspace history from browser IndexedDB (`PrismDB`):

- **1.2.1** Load relevant conversation history, saved preferences, and agent state.
- **1.2.2** Resolve referenced files, repositories, and available permissions.

1.3 Build a structured task record containing the goal, constraints, context, allowed tools, and expected output.

- **1.3.1** Record deadlines, output format, and other task constraints.
- **1.3.2** Attach the MCP tools and permissions required for execution.

1.4 Perform input boundary and security screening:

- **1.4.1** Check input size and screen for prompt-injection patterns.
- **1.4.2** Mark file, email, or other state-changing actions for user approval.

Step 2: Intent Classification and Capability Check

2.1 Classify the prompt into a functional domain such as automation, document work, research, or coding:

- **2.1.1** Extract semantic and keyword features from the prompt.
- **2.1.2** Use the intent classifier to select the most likely domain.

2.2 Profile required system capabilities and operational risk levels:

- **2.2.1** Identify required resources, such as web search, file access, or terminal execution.
- **2.2.2** Separate read-only work from actions that need approval.

2.3 Select optimal execution pathway:

- **2.3.1** Send simple requests directly to one model.
- **2.3.2** Send complex requests to the DAG planner for decomposition.

Step 3: DAG Decomposition and Model Routing

3.1 Deconstruct objective into Directed Acyclic Graph $\mathcal{G} = (V, E)$:

- **3.1.1** Create one vertex for each atomic task and define its expected output.
- **3.1.2** Connect tasks with directed edges when one task depends on another's output.
- **3.1.3** Hold a dependent task until all required inputs are available.

3.2 Compute topological task execution schedule $\pi = \mathrm{TopologicalSort}(\mathcal{G})$:

- **3.2.1** Find independent tasks that can run in parallel ($K \leq 3$).
- **3.2.2** Release dependent tasks when their parent tasks finish.

3.3 Select a model for each sub-task by balancing cost, latency, and quality:

$$\text{Provider}^* = \arg\min_m \left(\lambda_{\text{cost}} \, \text{Cost}_m + \lambda_{\text{latency}} \, \text{Latency}_m \right) \quad \text{s.t. } \text{Quality}_m \geq \tau$$

Meaning: Choose the lowest-cost and lowest-latency model that still meets the required quality threshold.

- **3.3.1** Use fast backends for simple or latency-sensitive tasks.
- **3.3.2** Use stronger models for reasoning, research, and code generation.

Step 4: Role-Based Task Assignment and Dispatching

4.1 Check the specialist-agent registry and their capability profiles:

- **4.1.1** Review each agent's role, instructions, and assigned tools.
- **4.1.2** Check worker availability before dispatch.

4.2 Match each sub-task to the agent with the closest capability profile:

$$\text{Agent}^{*} = \arg\max_{a \in \mathcal{A}} \ \text{CosineSimilarity}\big(\mathbf{Req}(v_i),\, \mathbf{Cap}(a)\big)$$

Meaning: Compare the task requirements with each agent's capabilities and choose the best match.

- **4.2.1** Send information-gathering tasks to the Researcher.
- **4.2.2** Send software tasks to the Coder.
- **4.2.3** Send checking and policy tasks to the Validator.

4.3 Manage task lifecycle states in dispatcher queue:

- **4.3.1** Keep tasks in `PENDING` until their dependencies finish.
- **4.3.2** Move ready tasks to `READY` and dispatch them.

Step 5: Agent Decision Cycle and Inter-Agent Communication

5.1 Run a ReAct cycle for each dispatched agent:

$$\text{State } s_t \longrightarrow \textbf{Observe} \longrightarrow \textbf{Reason} \longrightarrow \textbf{Act} \longrightarrow \textbf{Reflect}$$

Meaning: Observe the task, reason about the next step, act with a tool or response, and reflect on the result.

- **5.1.1** **Observe:** Read the task, constraints, and parent outputs.
- **5.1.2** **Reason:** Choose the next operation.
- **5.1.3** **Act:** Call an MCP tool or generate an answer.
- **5.1.4** **Reflect:** Check the result and correct local errors.

5.2 Transmit intermediate artifacts via DAG edge message passing:

- **5.2.1** Pass verified outputs from completed tasks to their dependent tasks.
- **5.2.2** Pass only relevant artifacts to limit context growth.

5.3 Synchronize shared state via local blackboard memory:

- **5.3.1** Save task state, conversation turns, and file references in `PrismDB`.
- **5.3.2** Notify dependent workers when a task completes.

Step 6: Specialist Execution Workflows – Planner & Assistant

6.1 Execute Planner Agent workflow (System Architect & Coordinator):

- **6.1.1** Read the objective and relevant context from `PrismDB`.
- **6.1.2** Split the objective into independent and dependent sub-goals.
- **6.1.3** Build the DAG and add capability tags.
- **6.1.4** Select models and enqueue the topological schedule.

6.2 Execute Assistant Agent workflow (Productivity & Communication):

- **6.2.1** Retrieve preferences and reference documents.
- **6.2.2** Create a structured draft or summary.
- **6.2.3** Use Google Workspace tools when required.
- **6.2.4** Verify the draft and request approval before sending it.

Step 7: Specialist Execution Workflows – Researcher, Coder & Validator

7.1 Execute Researcher Agent workflow (Information Retrieval & Synthesis):
- **7.1.1** Create search queries and use the Browser MCP server.
- **7.1.2** Extract relevant information from retrieved documents.
- **7.1.3** Remove duplicates and prepare a cited report.

7.2 Execute Coder Agent workflow (Software Synthesis & Execution):
- **7.2.1** Inspect the repository and edit the required source files.
- **7.2.2** Run tests or scripts in the terminal and inspect errors.
- **7.2.3** Fix syntax and runtime problems using test feedback.

7.3 Execute Validator Agent workflow (Quality Auditing & Safety Gate):
- **7.3.1** Check outputs against safety, format, and functional requirements.
- **7.3.2** Return **PASS** for valid output or **FAIL** with diagnostics.

Step 8: Standardized Tool Execution via Model Context Protocol

8.1 Interface agents to external tools via Model Context Protocol (MCP):

- **8.1.1** Use JSON-RPC for consistent client-server communication.
- **8.1.2** Discover tool schemas, inputs, and return types at runtime.
- **8.1.3** Keep agent logic independent of tool implementation details.

8.2 Orchestrate multi-domain tool execution ecosystem:

- **8.2.1** **System & Dev Tools:** Filesystem, terminal, and GitHub.
- **8.2.2** **Productivity Tools:** Gmail and Google Drive.
- **8.2.3** **Web Intelligence:** Browser search and document retrieval.

8.3 Enforce Human-in-the-Loop (HITL) authorization policy:

- **8.3.1** Allow read-only queries to run automatically.
- **8.3.2** Request approval for file changes, email sending, or other state changes.

Step 9: Context Compaction and Shared Memory Management

9.1 Address context window constraints and token consumption limits:

- **9.1.1** Keep the active context within the model's token limit.
- **9.1.2** Reduce unnecessary history to control latency and attention cost.

9.2 Compact the context by keeping verified outputs and removing stale or repeated content.

- **9.2.1** Forward only verified parent outputs to dependent tasks.
- **9.2.2** Remove redundant logs, drafts, and intermediate reasoning.

9.3 Persist global execution state using client-side database (`PrismDB`):

- **9.3.1** Save session turns, task graphs, and tool logs in IndexedDB.
- **9.3.2** Preserve local state for privacy and recovery after failure.

Step 10: Single-Stage Preference Alignment (ORPO) and Verification

10.1 Train the response-quality model with chosen and rejected examples using ORPO:

$$\mathcal{L}_{\text{ORPO}} = \mathcal{L}_{\text{SFT}} + \lambda_{\text{OR}} \cdot \mathcal{L}_{\text{OR}} \quad \text{where} \quad \mathcal{L}_{\text{OR}} = -\log \sigma \left(\log \left[\frac{\text{odds}_\theta(y_w \mid x)}{\text{odds}_\theta(y_l \mid x)} \right] \right)$$

Meaning: Combine normal supervised loss with a preference loss that favors the chosen response y_w over the rejected response y_l.

10.1.1 Increase the likelihood of chosen responses and reduce the likelihood of rejected responses.

10.1.2 Train on preference pairs (x, y_w, y_l) without a separate reward model.

10.2 Apply a quality gate before accepting an output:

10.2.1 Score each deliverable with the fine-tuned preference model.

10.2.2 Check instruction following, format, and safety requirements.

10.3 Handle quality failures via targeted DAG re-planning:

10.3.1 Record errors when an output fails the quality threshold.

10.3.2 Send the failed branch back to the planner for targeted revision.

Step 11: Artifact Assembly and Global State Commit

11.1 Aggregate and assemble multi-agent deliverables:

- **11.1.1** Collect verified outputs from completed DAG tasks.
- **11.1.2** Combine reports, code changes, and formatted content into one deliverable.

11.2 Evaluate the final deliverable for correctness, completeness, and timeliness:

- **11.2.1** Reject or revise the deliverable if a required dimension is weak.
- **11.2.2** Log quality results to improve future routing decisions.

11.3 Save the completed session state:

- **11.3.1** Write conversation records, DAG state, and tool receipts to `PrismDB`.
- **11.3.2** Record latency, token use, and audit information.

Step 12: Output Delivery and Reactive Workspace Presentation

12.1 Synthesize and package final deliverable artifacts:

- **12.1.1** Package reports, verified code patches, and document drafts.
- **12.1.2** Create an audit receipt with agent, timing, and tool information.

12.2 Render presentation layer via multi-pane workspace interface:

- **12.2.1** Show task states such as `Queued`, `Running`, `Success`, and `Failed`.
- **12.2.2** Render the result with formatting, syntax highlighting, and export controls.
- **12.2.3** Show approval controls for pending external actions.

12.3 Synchronize client state reactively via local database transactions:

- **12.3.1** Save verified state changes and output artifacts in `PrismDB`.
- **12.3.2** Update the interface when local database state changes.

Project Architecture, Working & System Design

Project workflow, architecture, working process, agent lifecycle, dataset pipeline, DAG scheduling, and flow diagrams

06. PrismSpace Workflow (Figure 1.0)

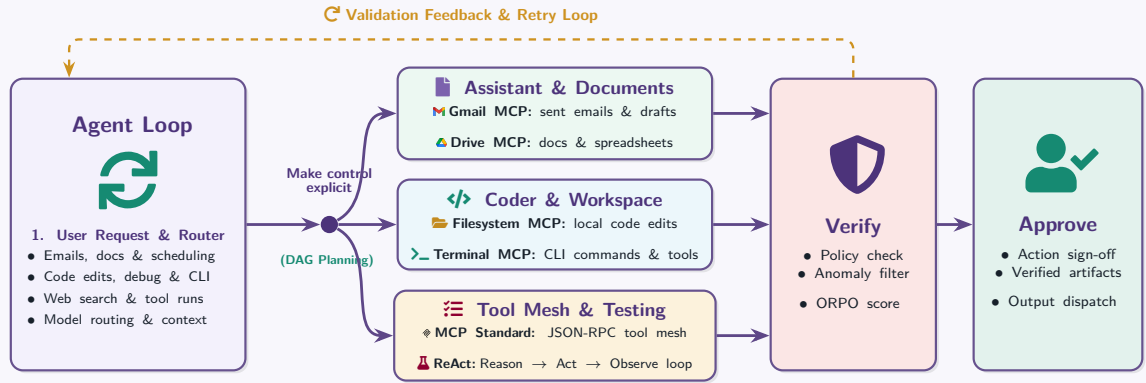


Figure 1.0: PrismSpace Workflow | **DAG** = Directed Acyclic Graph | **MCP** = Model Context Protocol | **ReAct** = Reason + Act | **ORPO** = Odds Ratio Preference Optimization | **HITL** = Human-in-the-Loop

10. Unified Dataset Preparation & Multi-Model Training Pipeline

Raw multi-source datasets flow through normalization and curation into four supervised tables, then train eleven production models.

1. Raw Datasets

- **WildGuardMix**
- **Agent trajectories**
- **RouterBench**
- **UltraFeedback**

`.parquet, .json, .jsonl`

normalize + curate

2. Supervised Tables: Fields & Training Roles

Dataset → Table	Training Fields	How Fields Are Used in Training
WildGuardMix `approval.jsonl`	`prompt`, `label (harm)`	Trains **ApprovalPredictor** & anomaly filter to block harmful/adversarial prompts.
Trajectories `success.jsonl`	`prompt`, `reward (0/1)`	Trains **SuccessPredictor** to estimate DAG task completion probability.
RouterBench `provider.jsonl`	`prompt, winner, vote_score`	Trains **ModelRouter** to select optimal provider (Groq vs OpenAI vs NIM).
UltraFeedback `reward.jsonl`	`prompt, chosen, rejected`	Trains **ORPO RewardModel** (Qwen2.5-1.5B) via preference ranking loss.

train + deploy

3. Production Models (11)

Routing & Intent Models (3)
- IntentClassifier
- AgentRouter
- ModelRouter: Groq / OpenAI / NIM

Workflow Intelligence & Performance (3)
- WorkflowClusterer
- SuccessPredictor
- LatencyPredictor

Safety, Alignment & Context (5)
- ApprovalPredictor, AnomalyDetector
- FAISS Vector Retriever
- RewardModel + Qwen2.5-1.5B ORPO

Dataset split: 80% training | 10% validation | 10% testing.
The held-out 20% is divided equally for validation and final testing.

10. Model Training Map: Algorithm Choice by Responsibility

Model-to-Algorithm Training Matrix

Model	Type	ML algorithm for training	Why choose this algorithm?
Intent Classifier	Classification	TF–IDF + XGBoost classifier	Fast and strong on sparse text features; captures nonlinear boundaries across the 12 intent classes.
Agent Router	Multilabel	Binary-relevance XGBoost	Predicts several specialist agents independently while preserving strong tabular/text-feature performance.
Approval Predictor	Classification	Calibrated XGBoost	Models nonlinear safety signals and produces probabilities that can be thresholded for HITL approval.
Workflow Success Predictor	Classification	XGBoost classifier	Learns interactions among task features, history, and plan shape; effective for structured trace data.
Provider Router	Classification	XGBoost ranking/classifier	Combines capability, quality, cost, and availability signals to choose the best provider/model.
Latency Predictor	Regression	XGBoost regressor	Captures nonlinear effects of provider load, prompt size, model, and tool complexity.
Cost Predictor	Regression	XGBoost regressor	Handles token-based and provider-specific cost interactions without requiring a linear cost assumption.
Anomaly Detector	Anomaly	Isolation Forest	Works with few or no anomaly labels and isolates rare, unusual execution patterns efficiently.
Reward ORPO Adapter	ORPO	ORPO + Reinforcement Learning	Learns from chosen-versus-rejected preference feedback to improve response quality and alignment.
FAISS Retrieval Index	Unsupervised	Sentence embeddings + FAISS	Requires no class labels; semantic vectors support fast nearest-neighbor retrieval over workspace context.
Workflow Clusters	Unsupervised	K-Means clustering	Groups similar execution traces to discover reusable workflow templates without labeled outcomes.

09. Model Responsibilities & Inference Flow

Intent: The **underlying goal** of a user's request. Determines *task type* (research, coding, tool use, synthesis) for routing to agents & tools.

1. Intent Classifier	**Prompt** → **Extracted features** → **classify intent** (research, tool use, coding, everyday synthesis)	**Task type** `intent_classifier.joblib`
2. Agent Router	**Intent + task features** → predict **multiple agent labels** → form the **specialist team**	**Agent selection** `agent_router.joblib`
3. Workflow Planner	**Workflow templates** → estimate **completion probability** → choose a **dependency-safe DAG plan**	**Plan reuse + success** `workflow_templates.pkl` `workflow_success_predictor.joblib`
4. Provider Routing	**Candidate task** → choose **provider/model** → compare **predicted latency** and **token cost**	**Provider + cost balance** `model_router.joblib` `latency_predictor.joblib` `cost_predictor.joblib`
5. Safety Gate	**Planned action** → score **approval risk** + **detect anomalies** → **approve, pause, or block** execution	**Risk approval & safety** `approval_predictor.joblib` `anomaly_detector.joblib`
6. Context & Quality	**Query** → **FAISS semantic retrieval** → add **context** → generate & **ORPO-align response**	**Evidence + output quality** `faiss.index` · `faiss.vectorizer.joblib` `reward_orpo_v2/adapter_model.safetensors`

Intent Classifier: Prompt Decomposition, NLP Algorithm & Serialization

Formal Pipeline: $\mathcal{T}_{\text{user}} \xrightarrow{\text{Dense Tokenize}} \mathbf{e} \in \mathbb{R}^{384} \xrightarrow{\text{Intent Classifier}} \hat{\mathbf{y}}_{\text{intent}} \xrightarrow{\text{Semantic Decomposition}} \mathcal{G}(V, E) \xrightarrow{\text{JSON Schema}}$ JSON-RPC Task Packet

Raw Input $\mathcal{T}_{\text{user}}$

"Find the latest API changes, summarize them, and update my notes."

vector $\mathbf{e}$

Intent Classifier

- **Extracts intent:** Understands requested user actions.
- **Maps task domains:** Identifies web search, synthesis & tools.
- **Enforces safety rules:** Flags file modifications for approval.

intents $\hat{\mathbf{y}}$

Semantic Role Decomposer

- **Algorithm:** Action-Entity-Precondition SRL
- **Decomposition:** 4 Vertices (V), 3 Edges (E)
- **Rule:** Causal pipeline $v_i \rightarrow v_{i+1}$

Node v_1: Research
SearchWeb("API changes")
Browser MCP · raw_docs

raw_docs

Node v_2: Extraction
ParseDOMDiff(raw_docs)
Parser MCP · api_diff

api_diff

Node v_3: Synthesis
Summarize(api_diff)
Writer Agent · summary_md

summary

Node v_4: Approval & Write
Filesystem.Write(summary)
HITL Gate $\Longrightarrow$ Commit notes

serialize

Strongly-Typed JSON-RPC Task Packet Schema

```
{
 "task_id":  "dag_8921", "algorithm":  "kahn_topological",
 "vertices":  [
  {"id":  "v1", "agent":  "Researcher", "mcp":  "browser.search", "deps":  []},
  {"id":  "v2", "agent":  "Analyst", "mcp":  "parser.extract", "deps":  ["v1"]},
  {"id":  "v3", "agent":  "Writer", "mcp":  "llm.summarize", "deps":  ["v2"]},
  {"id":  "v4", "agent":  "Filesystem", "mcp":  "fs.write_file", "deps":  ["v3"],
"approval_gate":  true}
 ]

}
```

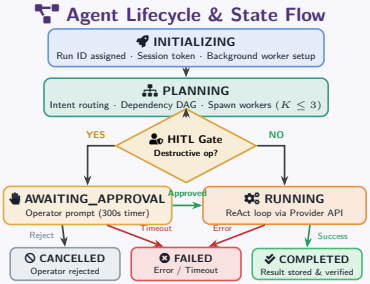
execute

Task Dispatcher

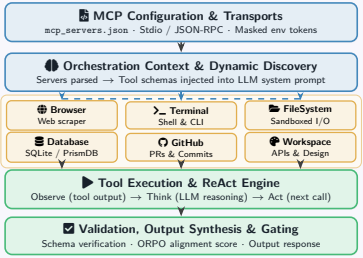
- **Runs tasks in order:** Dispatches ready tasks and passes outputs forward.
- **Enforces approval:** Pauses before writing files until user confirms.
- **Verifies quality:** Checks outputs before saving results.

07. Agent Lifecycle, MCP Integration & Runtime Flow in PrismSpace

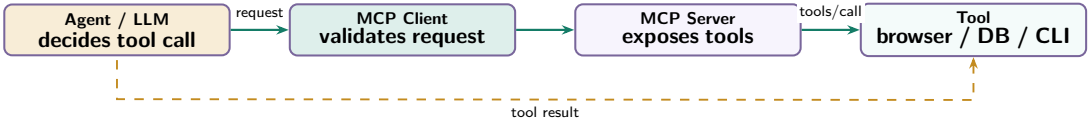
Agent Lifecycle & State Flow



MCP Tool Integration & Runtime Flow



How Agents Communicate with MCP




```
-> request:  {"jsonrpc":"2.0","id":7,"method":"tools/call","params":{"name":"search","arguments":{"q":"MCP"}}}
<- response:  {"jsonrpc":"2.0","id":7,"result":{"content":[{"type":"text","text":"3 matches"}]}}
```

11. How the System Works: From Request to Result

Stage 1: Ingestion & Features

- **Capture request:** Parse user input (text/voice/API).
- **Clean & normalize:** Remove noise, handle missing data with NLP preprocessing.
- **Extract features:** Tokenize into semantic embeddings using transformer models.

Stage 2: Intelligence & DAG

- **Classify intent:** ML classifier identifies task type from 12 categories (99.6% accuracy).
- **Choose agents & models:** Route sub-tasks to specialized LLM agents.
- **Topological sorting:** Order DAG tasks by dependencies; execute only after all prerequisites finish.

Stage 3: Execution & Storage

- **Execute:** Run up to 3 parallel tasks using Model Context Protocol (MCP) tools.
- **Track progress:** Real-time telemetry, node state transitions, and execution logs.
- **Verify & store:** ORPO quality check, persist results to PrismDB.

Building & Executing the Task Graph $\mathcal{G}(V, E)$

1. User Goal Prompt
"Analyze news+CSV, email summary"

→

2. Split into Vertices V
Modular async tasks A, B, C, D

→

3. Connect Edges E
Map output return → input args

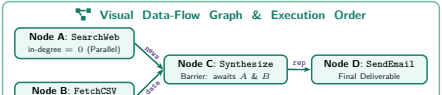
→

4. Topological Sort
Pick in-degree $= 0$; unlock dependents

Computer Science Explanation (By Concrete Example):

- **Vertices (V) = Sub-Task Functions (Async Call Units):**
  Node A: `news = await SearchWeb("Tesla news")` *[Researcher]*
  Node B: `data = await FetchCSV("metrics.csv")` *[Data Agent]*
  Node C: `rep = await Synthesize(news, data)` *[Writer Agent]*
  Node D: `await SendEmail(to="lead", body=rep)` *[Email Tool]*
- **Edges (E) = Data Pipes & Synchronization Barriers:**
  Edge (A, C) pipes return value `news`; edge (B, C) pipes `data`.
  Barrier: task C blocks until both promises A & B resolve.
- **Connection Basis:** C links to A & B because its inputs require their outputs; A and B are independent.
- **Topological Sorting (Kahn's method):** repeatedly select nodes with in-degree $= 0$, remove outgoing edges, and unlock dependents.
  This gives a valid dependency order and exposes independent work for parallel execution.
- **Execution:** A and B run in parallel; after both finish, C unlocks, then D runs.

Visual Data-Flow Graph & Execution Order



Parallel Thread 1: Worker 1 executes Node A → returns `news`
Parallel Thread 2: Worker 2 executes Node B → returns `data`
Join Barrier: Node C consumes `news` + `data` → outputs `rep`

Serial Finish: Node D consumes `rep` → sends email

Why Topological Sorting is Chosen for DAG Task Scheduling

1. Partial Order Precedence

- **Causal constraints:** Task v blocks until inputs $u \in \text{Pred}(v)$ finish.
- **No false priority:** Independent tasks (A, B) share no forced order.

2. Finds Cycles & Prevents Hangs

- **Finds circular dependencies:** Detects when tasks wait for each other in a loop $(A \to B \to A)$.
- **Stops safely:** Reports the problem instead of waiting forever for tasks to finish.

3. Parallel Concurrency Fan-out

- **Zero in-degree queue:** Nodes with in-degree $= 0$ run **concurrently**.
- **Barrier sync:** Automatically resolves dependencies into join barriers.

Comparative Analysis: Topological Sort vs. Classical Sorting Algorithms

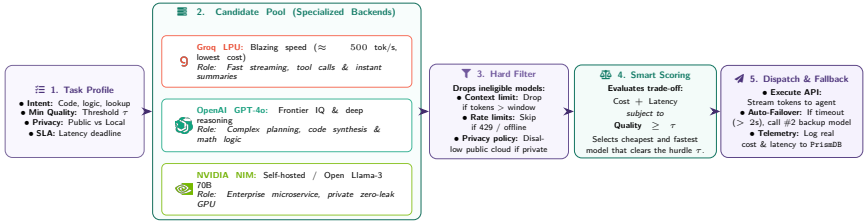
Algorithm	Data Structure	Ordering Principle	Time Compl.	Cycle Detect	Parallelism	Fit for Workflow DAGs
Topological Sort (Kahn's / DFS)	Directed Acyclic Graph $\mathcal{G}(V, E)$	**Causal Precedence** (Partial Order: $u \to v$)	$\mathcal{O}(V + E)$	Yes (In-degree)	Yes (Fan-out)	**Optimal:** Respects data pipes; runs async independent tasks in parallel.
Merge Sort (Divide & Conquer)	1D Array / Linear List	**Value Magnitude** (Total Order: $a \leq b$)	$\mathcal{O}(n \log n)$	No (N/A)	No (Serial 1D)	**Invalid:** Ignores dependency graphs; compares data values, not prerequisites.
Quick Sort (Pivot Partitioning)	1D Array / Linear List	**Value Magnitude** (Total Order: $a \leq b$)	$\mathcal{O}(n \log n)$ avg	No (N/A)	No (Serial 1D)	**Invalid:** Partitions around arbitrary value pivots; destroys execution order.
Heap Sort (Priority Queue)	Binary Tree / Array	**Key Priority** (Total Order: max/min)	$\mathcal{O}(n \log n)$	No (N/A)	No (Serial 1D)	**Invalid:** Orders by scalar priority; cannot enforce multi-parent edge barriers.
Bubble / Insertion (Linear Swap)	1D Array / Linear List	**Adjacent Swap** (Total Order: $a_i > a_{i+1}$)	$\mathcal{O}(n^2)$	No (N/A)	No (Strict Serial)	**Invalid:** Quadratic pairwise swaps cannot represent topological DAG flow.

Core Architectural Takeaway: Classical sorting algorithms ($\mathcal{O}(n \log n)$) arrange scalar values along a 1D axis ($a_1 \leq a_2 \ldots$). In contrast, multi-agent execution graphs require scheduling a **partially ordered set** with zero, one, or multiple asynchronous parent promises. **Topological Sorting** is the only algorithm with $\mathcal{O}(V + E)$ complexity that guarantees causal precedence, detects deadlocks, and unlocks parallel execution.

How Provider Routing Works: Task-to-Model Selection

Core Routing Mission: Instead of sending every query to a slow, expensive frontier model, PrismSpace dynamically routes each agent sub-task to the **fastest, most cost-effective provider** that clears the task's required **quality threshold (τ)**.

End-to-End Provider Routing Pipeline



How the Router Decides (Multi-Objective Optimization)

Objective: $\min_m \left(\lambda_c \cdot \widehat{\text{Cost}}_m + \lambda_l \cdot \widehat{\text{Latency}}_m \right)$ s.t. $\widehat{\text{Quality}}_m \geq \tau$

- **Speed-Critical Nodes ($\lambda_l \uparrow$):** Fast tool calls prioritize low latency $\rightarrow$ routes to **Groq LPU** (< 300ms).
- **High-Complexity Tasks ($\tau \uparrow$):** Architectural synthesis prioritizes IQ $\rightarrow$ routes to **OpenAI GPT-4o**.
- **Confidential Workflows:** Data protection requirements lock execution to local **NVIDIA NIM** microservices.

Runtime Safeguards & Zero-Downtime Fallback

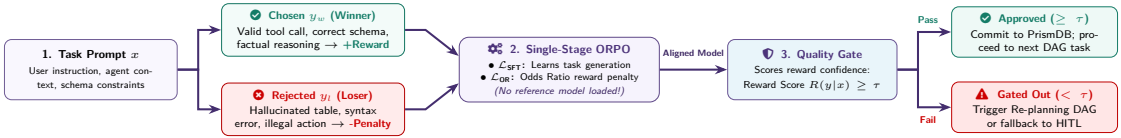
- **Pre-Flight Verification:** Validates endpoint health, remaining quota, and network jitter before dispatching tokens.
- **Dynamic Circuit Breaker:** If the primary provider times out (> 2000ms) or hits HTTP 429/503 rate limits, execution instantly cascades to the #2 ranked model (< 150ms failover).
- **Closed-Loop Feedback:** Logs observed token speed, dollar cost, and quality scores to continuously calibrate future routing models.

How ORPO Works: Preference Alignment and Quality Gating

What is ORPO & Why is it Used Here as a Reward System?

- **What is ORPO? Odds Ratio Preference Optimization (ORPO)** is an integrated **single-stage reward and alignment algorithm**. Traditional RLHF (PPO) requires training a separate reward model and keeping a frozen reference model in GPU memory (heavy, slow, and unstable).
- **Why in PrismSpace?** In multi-agent swarms, LLMs can hallucinate invalid tools or malformed code. ORPO acts as an **internal reward filter**: it mathematically rewards compliant trajectories (y_w) and penalizes bad responses (y_l) during fine-tuning, eliminating the need for an external reward model.

End-to-End Pipeline: From Preference Pairs to Runtime Quality Gating



ORPO vs. Traditional RLHF

- **The RLHF (Reinforcement Learning from Human Feedback) Problem:** Traditional alignment (PPO/DPO) requires running 3 separate models simultaneously: Policy Model, Reference Model, and Reward Critic. This demands > 48 GB VRAM and suffers from unstable policy gradients.
- **The ORPO Breakthrough:** It mathematically merges preference alignment directly into supervised fine-tuning in a **single stage**.
- **Concrete Benefit: 50% GPU memory savings**, faster convergence, and no policy drift without loading any secondary reference model!

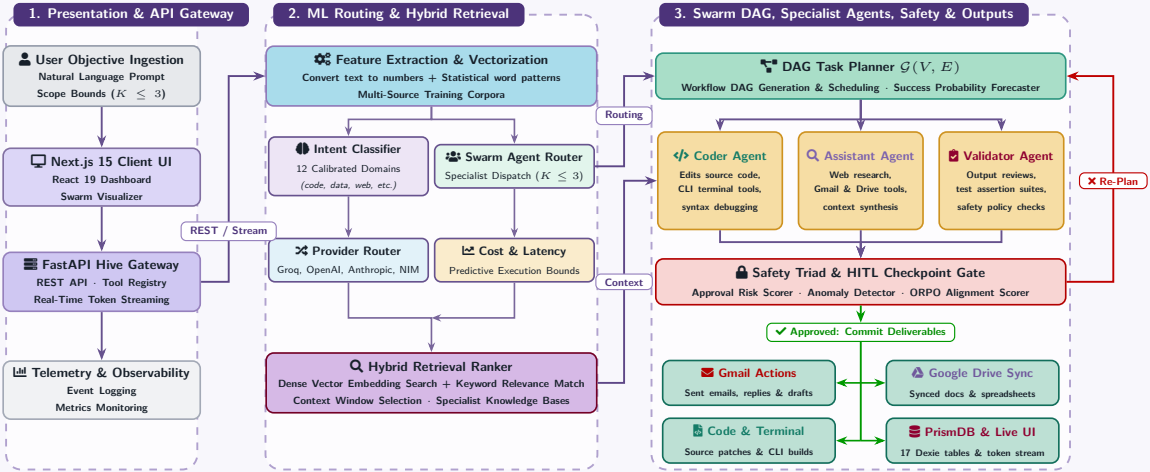
The Mathematical Reward Intuition

$$\mathcal{L}_{\text{ORPO}} = \mathcal{L}_{\text{SFT}} + \lambda_{\text{OR}}\, \mathcal{L}_{\text{OR}}$$

$$\mathcal{L}_{\text{OR}} = -\log \sigma\left(\log \frac{\text{odds}(y_w|x)}{\text{odds}(y_l|x)}\right), \quad \text{odds}(y|x) = \frac{P(y|x)}{1-P(y|x)}$$

- **Implicit Reward Gradient:** $\text{odds}(y|x)$ measures the probability of generating a token vs not generating it. When y_l is chosen, loss explodes, forcing logits of $y_w \gg y_l$.
- **Runtime Impact:** The model naturally generates schema-compliant tool calls and suppresses hallucinations before writing to PrismDB.

12. PrismSpace System Architecture & Multi-Model Swarm Workflow (Figure 1.2)



13. Flowchart: Multi-Agent Execution Flow (Figure 1.3)

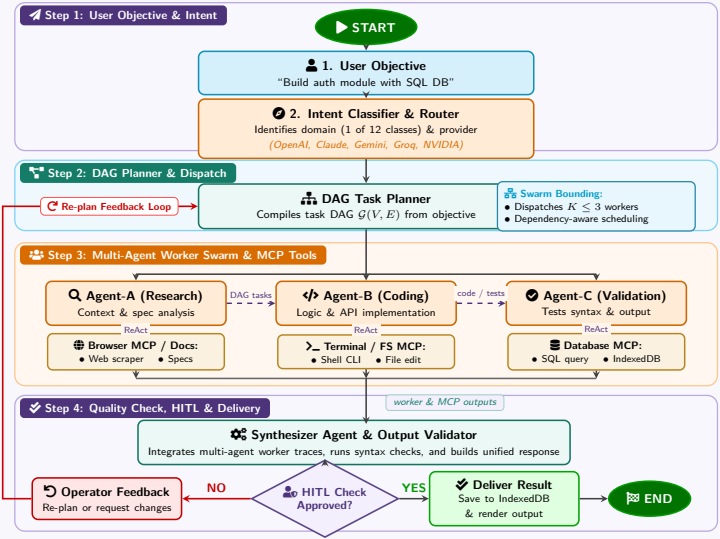
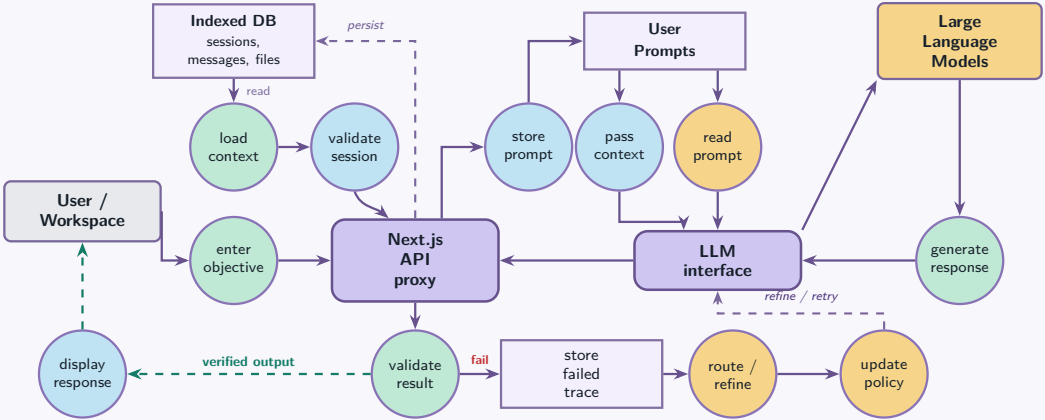


Figure 1.3: Simplified Multi-Agent Communication Flow in PrismSpace

07. PrismSpace Data Flow Diagram



Database Design

PrismDB schema, relationships, and entity–relationship structure

14. Database Schema

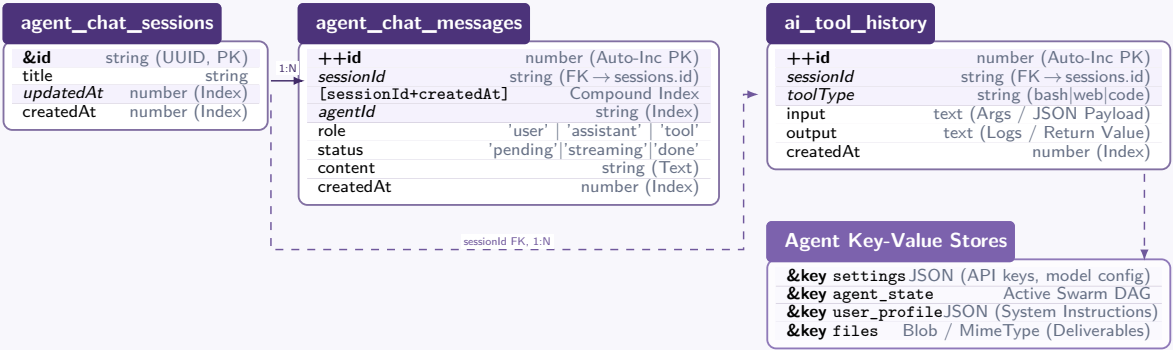
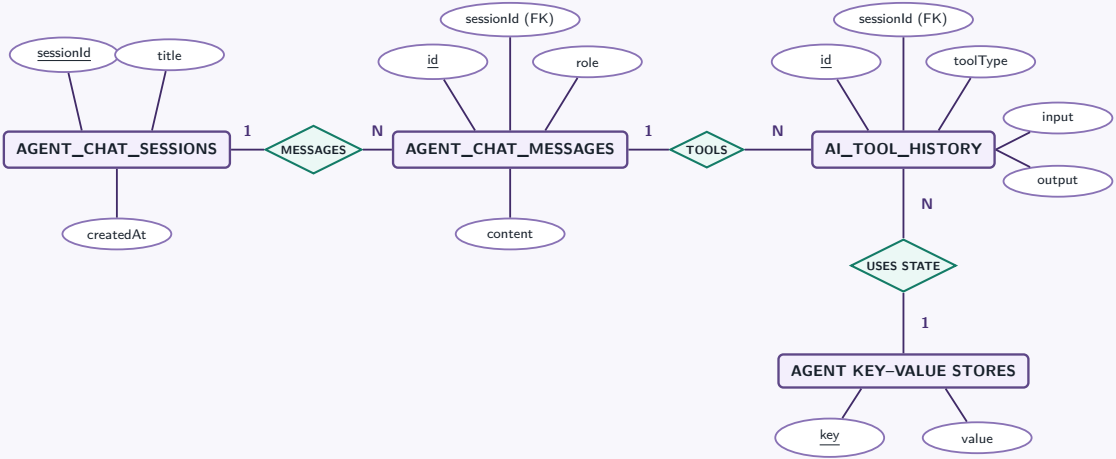


Figure 1.4: PrismDB - AI-Agent Browser IndexedDB Schema (Dexie.js v3) | **PK**: Primary Key **FK**: Foreign Key **Index**: B-tree index
`sessions.id` (Shared FK) → `messages.sessionId` & `tool_history.sessionId` | `useLiveQuery()` Reactive UI | ACID Dexie.js

15. Database ER Diagram



Project Requirements

Hardware and software needed to build and run PrismSpace

16. Hardware Requirements

- **CPU:** Multi-core processor for the Next.js client, FastAPI service, and local development tools.
- **GPU:** NVIDIA RTX 3050 or above for embedding generation, vector indexing, and model-training workloads.
- **Memory:** Minimum 16 GB RAM for running the frontend, backend, databases, and supporting services concurrently.
- **Storage:** SSD with sufficient capacity for the codebase, PrismDB data, logs, model checkpoints, and FAISS indexes.
- **Network:** Reliable internet connection for provider APIs, MCP integrations, and model downloads.
- **Security:** Secure environment-variable or secret-manager support for provider and tool credentials.

17. Software Requirements

- **Next.js 14 / React / TypeScript:** browser-based workspace for objectives, chat history, agent status, and approvals.
- **FastAPI / Python:** asynchronous backend framework for agent orchestration, REST endpoints, and runtime events.
- **Dexie.js / IndexedDB:** local ACID storage for sessions, messages, files, settings, and tool history.
- **Joblib / scikit-learn:** Python libraries for data preparation, utility functions, serialization, and general machine-learning support.
- **MCP JSON-RPC:** MCP (Model Context Protocol) uses JSON-RPC (JavaScript Object Notation–Remote Procedure Call) messages to standardize tool names, inputs, outputs, and errors for browser, terminal, filesystem, GitHub, database, and workspace tools.
- **Provider SDKs/APIs: NVIDIA NIM** provides optimized model-serving endpoints; the **Groq SDK** supports low-latency inference; and the **Anthropic Claude SDK/API**, OpenAI-compatible APIs, and similar provider clients connect the orchestrator to different language models.

Project Implementation

Frontend, backend, and training components

18. Frontend Implementation

- **Next.js 14:** React-based framework for building the browser application and managing frontend pages.
- **React:** component library for creating reusable user-interface elements.
- **TypeScript:** typed JavaScript used for safer and more maintainable frontend development.
- **Tailwind CSS:** utility-first styling framework for responsive layouts and consistent visual design.
- **Dexie.js, IndexedDB, and dexie-react-hooks:** browser storage and reactive data-access libraries for local application data.
- **Lucide React:** icon library used for interface symbols and controls.
- **Framer Motion and GSAP (GreenSock Animation Platform):** frontend animation libraries for smooth interface transitions.

19. Backend Implementation

- **FastAPI:** Python framework for building asynchronous backend services and application endpoints.
- **Python:** primary backend language for API services, integrations, data handling, and runtime coordination.
- **Uvicorn:** lightweight **ASGI (Asynchronous Server Gateway Interface)** server used to run FastAPI. ASGI connects asynchronous Python applications to web servers and supports non-blocking concurrent requests.
- **REST (Representational State Transfer):** API style for communicating with the frontend through structured HTTP requests and responses.
- **Pydantic:** schema validation and typed data models for request, response, configuration, and tool payloads.
- **MCP (Model Context Protocol) and JSON-RPC (JavaScript Object Notation–Remote Procedure Call):** standards for structured communication between the backend and external tools.
- **Provider SDKs/APIs:** integration layer for connecting the backend to external language-model and model-serving services.

20. Model Training Modules

- **Dataset preparation:** collects, cleans, and combines task traces and preference data using **Pandas**, **NumPy**, and **PyArrow**.
- **Intent classification:** categorizes user objectives using **scikit-learn** text features, classifiers, and **XGBoost**.
- **Agent routing:** selects specialist agents using shared **scikit-learn** pipelines and **XGBoost** classification support.
- **Model and provider routing:** compares task features and provider signals using **Pandas**, **NumPy**, **scikit-learn**, and **XGBoost**.
- **Performance prediction:** estimates latency and cost with **XGBoost** regression, supported by **scikit-learn** preprocessing.
- **Safety prediction:** estimates approval needs and detects unusual patterns with **scikit-learn**, **XGBoost**, and **Isolation Forest** utilities.
- **Retrieval:** uses **FAISS (Facebook AI Similarity Search)**, **NumPy**, and optional **sentence-transformers** embeddings for context search.
- **Preference alignment:** uses **PyTorch**, **Hugging Face Transformers**, **TRL (Transformer Reinforcement Learning)**, **PEFT**, and **Datasets** for ORPO training.
- **Runtime artifacts:** uses **Joblib**, **Safetensors**, and **Hugging Face Accelerate** to save, validate, and load trained components.

Artifact: a saved output from training, such as a trained model, vector index, or configuration file, that can be reused later. **FAISS (Facebook AI Similarity Search):** a library for quickly finding similar vectors. **PEFT (Parameter-Efficient Fine-Tuning):** adapts a model by training only a small set of additional parameters.

System Design

Modules, responsibilities, and system-level organization

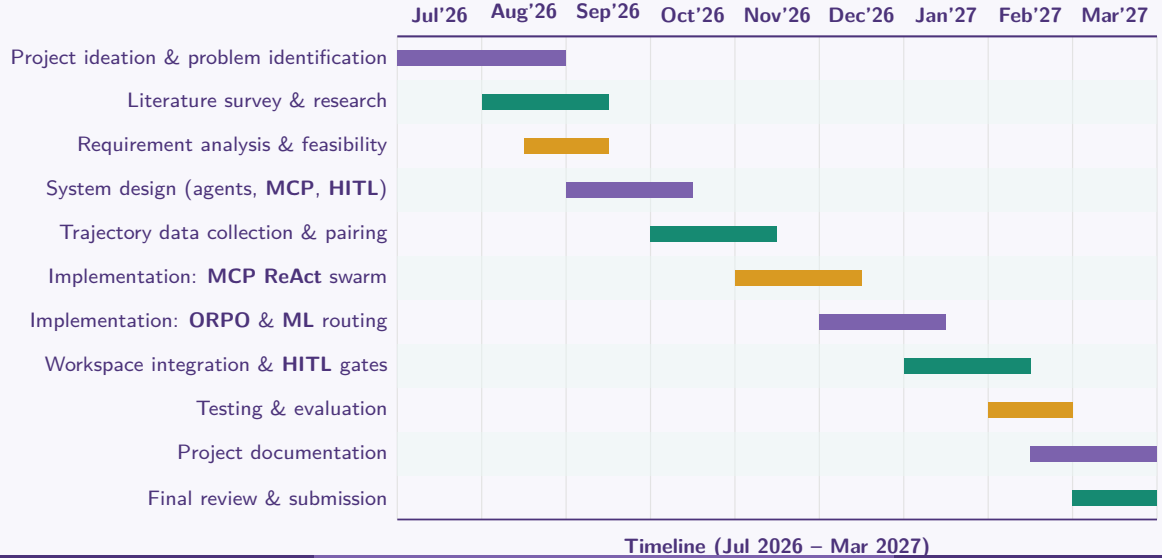
21. System Modules and Responsibilities

- **User and Session Module:** captures objectives, settings, conversation history, uploaded files, and session state.
- **Context and Routing Module:** classifies intent and selects the most suitable provider, model, and specialist agents.
- **DAG (Directed Acyclic Graph) Planning Module:** decomposes objectives into dependency-aware subtasks and schedules parallel work.
- **Agent Execution Module:** runs bounded **ReAct (Reasoning and Acting)** loops and invokes approved tools.
- **MCP (Model Context Protocol) Integration Module:** exposes filesystem, terminal, database, memory, and external-service tools through **JSON-RPC (JavaScript Object Notation–Remote Procedure Call)**.
- **Verification and Safety Module:** applies schema checks, **ORPO (Odds Ratio Preference Optimization)** preference gating, anomaly detection, and **HITL (Human-in-the-Loop)** approvals.
- **Persistence Module:** stores sessions, messages, artifacts, and errors in the local **PrismDB (browser IndexedDB database)**.

22. Conclusion

- **PrismSpace** converts a user's natural-language goal into smaller, organized tasks.
- Independent tasks can be executed by multiple agents in parallel.
- Standard tools are connected through the **Model Context Protocol (MCP)**.
- Human approval and local browser storage improve safety, privacy, and control.
- The platform provides a flexible foundation for future agent-based applications.

23. Gantt Chart



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Thank You!

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